CA Note

1 Basic Knowledge

Group $(G,*)$**:** **G1**: Closure **G2**: Associativity **G3**: Identity element **G4**: Inverse element $\downarrow$ **Division ring**: $R^\times = R\setminus\{0\}$; $R^\times$ group (under $\cdot$)

Ring $(R,+,\cdot)$**:** **R1**: $(R,+)$ is an abelian group **R2**: $(R,\cdot)$ is monoid. (no inverse) **R3**: Distributive law (left and right)

Ideal (I)**:** **I1:** $\neq\emptyset$ **I2:** $\forall a,b\in I, a-b\in I$ **I3:** $\forall r\in R, i\in I, ri, ir\in I$ **Lemma:** $I=R \Leftrightarrow 1\in I \Leftrightarrow I\cap R^\times \neq \emptyset$

Subgroup (H): **H1**: $e\in H$ **H2**: $\forall a,b\in H, ab\in H$ **H3**: $\forall a\in H, a^{-1}\in H$ **Subring** (S): **S1**: $1\in S$ **S2**: $\forall a,b\in S, a-b\in S$ **S3**: $\forall a,b\in S, ab\in S$

Integral Domain (ID): ID1: communicative ring **ID2**: no zero divisors **Lemma:** Finite ID $\Rightarrow$ field ID $\Rightarrow$ Cancellation Law

Factor Ring (R/I)**:** $a+I=b+I \Leftrightarrow a-b\in I$ $(R/I,\oplus,\odot)$ is a ring. ($1_{R/I}=1_R+I, 0_{R/I}=0_R+I$, I ideal) $\mathbb{Z}_n\cong\mathbb{Z}/n\mathbb{Z}$

Ring Homomorphism: $\varphi: R\to S$ s.t. **RH1**: $\varphi(a+b)=\varphi(a)+\varphi(b)$ **RH2**: $\varphi(ab)=\varphi(a)\varphi(b)$ **RH3**: $\varphi(1_R)=1_S$

· **1**: Ker(φ) is an ideal. **2**: Im(φ) is a subring. **3**: φ is 1-1 $\Leftrightarrow$ Ker$(\varphi)=\{0_R\}$. **FIT:** $R/\operatorname{Ker}(\varphi)\cong \operatorname{Im}(\varphi)$ (First Isomorphism Theorem)

· **Canonical Homomorphism:** $can: R\to R/I$ s.t. $can(r)=r+I$ is a onto ring Homomorphism. Ker$(can)=I$ $\updownarrow$ ($\exists!\tilde\varphi$ 双射)

· **Lemma:** If $\varphi: R\to S$ ring Homomorphism s.t. $I\subseteq \operatorname{Ker}(\varphi)$, then $\exists!\tilde\varphi: R/I\to S$ s.t. $\varphi=\tilde\varphi\circ can$ (i.e. $\tilde\varphi(r+I)=\varphi(r)$)

Lagrange's Theorem: G finite group, $H\le G$ subgroup, then $|H|\mid|G|$

2 Types of Ideals **(Maximal Ideals $\Rightarrow$ Prime Ideals $\Rightarrow$ Radical Ideals $\Rightarrow$ Ideals)**

Division: If R ring, then: $\forall f,g\in R[x]$, g 首相系数可逆 $\Rightarrow$ $\exists q,r\in R[x]$ **s.t.** $f(x)=g(x)q(x)+r(x)$ **and** $\deg(r)<\deg(g)$ **or** $r=0$.

Lemma: If R ring, $g\in R[x], g\neq 0, I=(g)$. $\Rightarrow$ [1] $\forall i\in R[x]/I, i=f+I, \deg(f)<\deg(g)$
g 首相系数可逆 [2] If $a,b\in R[x], \deg(a,b)<\deg(g)$ and $a+I=b+I$, then $a=b$. (系数也相同)

Remark: If $I=(g)$, g 首相系数不可逆: Find a polynomial g_1 s.t. $g_1=hg, h\in R[x]$ 首相系数可逆, 那么可满足 [1] 条件,[2] 只能 subset 但不能唯一表达.

Remark: 有些时候我们能用 gcd, lcm 计算 $<a,b>=<\gcd(a,b)>$ 和 $a\cap b=<\operatorname{lcm}(a,b)>$ [If R communicative ring] 但注意! 要在能够用 Division Algorithm | 有 gcd, lcm 的环里用.

Principal Ideal: Ideal I: Iff $I=aR$ **Prime Ideal:** Ideal I≠R: $ab\in I\Rightarrow a\in I$ or $b\in I$. (OR: If $a\notin I$ and $b\notin I$, then $ab\notin I$)
Maximal Ideal: Ideal I≠R: If J ideal and $I\subseteq J\subseteq R$, then $J=I$ or $J=R$. $\downarrow$ I not radical $\Leftrightarrow \exists a\notin I$ s.t. $a^n\in I$ for some $n\in\mathbb{N}$
Radical|Radical Ideal: I≠R: **Radical**: $\sqrt{I}=\{r\in R: r^n\in I \text{ for some } n\in\mathbb{N}\}$ **Radical Ideal**: If $I=\sqrt{I}$ $I\subseteq\sqrt{I}$ always.

Properties: Relationship between these kinds of ideals:

1. R communicative ring, I ideal. **I prime $\Leftrightarrow R/I$ is an ID.** **I maximal $\Leftrightarrow R/I$ is a field**
2. R ring, $I_1,I_2,\dots,I_n\lhd R$, $I_1\cap I_2\cap\cdots\cap I_n=\{0\}$ $\Rightarrow$ R isomorphic to a subring of $R/I_1\oplus R/I_2\oplus\cdots\oplus R/I_n$. $R\cong R/I_1\oplus R/I_2\oplus\cdots\oplus R/I_n$ if $I_i+I_j=R$ for $i\neq j$
3. R ring, I ideal, J ideal of R/I, $T=\{r\in R: r+I\in J\}$(reps of J). Then [1] **T: Ideal of R** [2] $I\subseteq T$ [3] $J=\{t+I: t\in T\}$ [4] $(R/I)/J\cong R/T$
   If $J=\langle j_1+I,\dots,j_n+I\rangle$ then $T=\langle j_1,\dots,j_n\rangle+I=\langle j_1,\dots,j_n,I\rangle$ Moreover, if $I=\langle i_1,\dots,i_m\rangle$, then $T=\langle j_1,\dots,j_n,i_1,\dots,i_m\rangle$
4. R communicative ring, I ideal. **I radical $\Leftrightarrow R/I$ has no nonzero nilpotent elements** except 0 (nilpotent: $r^n=0$)
5. If R (commutative with *unity*), then **any ideal $I\neq R$ is contained in a *maximal* ideal.** If r is *not a unit*, then $1\notin\langle r\rangle=rR$

3 PID | UFD | Euclidean Domain **(ED $\Rightarrow$ PID $\Rightarrow$ UFD $\Rightarrow$ ID)**

Irreducible: $0\neq a\in R\setminus R^\times$ is **Irreducible**: If $a=bc$, then $b\in R^\times$ or $c\in R^\times$.
Prime: $0\neq a\in R\setminus R^\times$ is **Prime**: If $a\mid bc$, then $a\mid b$ or $a\mid c$. (If R communicative ring, then prime $\Rightarrow$ irreducible)

1. F field, $0\neq g\in F[x]$ non-constant: [1] **$F[x]/\langle g\rangle$ is a field** $\Leftrightarrow$ [2] **g is irreducible in $F[x]$** $\Leftrightarrow$ [3] **$F[x]/\langle g\rangle$ is a domain**
2. F field, $0\neq g\in F[x]$: **g is irreducible $\Leftrightarrow$ $\langle g\rangle$ is prime ideal**
3. If R is ID: **Prime $\Rightarrow$ Irreducible** If $0\neq a\in R$, then: **a prime $\Leftrightarrow aR$ prime ideal**
4. If R is PID, $a\neq 0$: [1] **aR is prime ideal** $\Leftrightarrow$ [2] **a is prime** $\Leftrightarrow$ [3] **aR is maximal ideal** $\Rightarrow$ [4] **a is irreducible**

PID: R communicative ring is *principal ideal domain* if: $\forall I\lhd R, I=aR$ for some $a\in R$. Theorem: **$\mathbb{Z}, F, F[x]$ is PID.**
ED: R is *ID*. R is *Euclidean Domain* if: **(Euclidean func)** $\exists v: R\setminus\{0\}\to\mathbb{N}_0$ s.t. $\forall a,b\in R, b\neq 0, \exists q,r\in R$ s.t. $a=bq+r$ and $r=0$ or $v(r)<v(b)$.
Common ED: $\mathbb{Z}, F[x], \mathbb{Z}[i]$Gaussian Integers are ED, with $v(a)=|a|, v(f)=\deg(f), v(a+bi)=a^2+b^2$ respectively.
Euclidean Algorithm: If R ED, $a,b\in R\setminus\{0\}$, then $\gcd(a,b)=xa+yb$ for some $x,y\in R$. (extended of Bezout's Identity)
UFD (Def 1): R ID is *unique factorization domain* if: **UFD 1**: $\forall a\in R\setminus\{0\}$, $\exists$ irreducibles $p_1,\dots,p_n\in R$ s.t. $a=p_1p_2\dots p_n$
UFD 2: Any other irreducibles $q_1,\dots,q_m\in R$ s.t. $a=q_1q_2\dots q_m$, then $n=m$ and $p_i=u_iq_i$ for some $u_i\in R^\times$ after reordering.
Acending Chain Condition (ACC): A sequence of ideals $I_1\subseteq I_2\subseteq I_3\subseteq\dots$ *satisfies ACC* if: $\exists n\in\mathbb{N}$ s.t. $I_n=I_{n+1}=I_{n+2}=\dots$
UFD (Def 2): R ID is *UFD* if: **UFD 1**: Every sequence of *Principal Ideals* satisfies Ascending Chain Condition (ACC)
UFD 2: If a irreducible, then a is prime ($\Leftrightarrow aR$ is a prime ideal.) **Causs's Theorem: If R is UFD, then $R[x]$ is UFD.**

1. **Bezout's Identity**: R is *PID*, then $\forall a,b\in R, \exists x,y\in R$ s.t. $ax+by=\gcd(a,b)$.
2. R UFD: **a prime $\Leftrightarrow a$ irreducible** · **R PID: Every sequence of ideals satisfies ACC.**

4 Localisation

Field of Fractions (F_R)**:** R is ID, $S=R\setminus\{0\}$. Def *equivalence relation* $\sim$ on $R\times S$ by: $(a,b)\sim(c,d)\Leftrightarrow ad=bc$. (Like $\frac{a}{b}=\frac{c}{d}$)
Then the *field of fractions* F_R of R: $F_R=(R\times S)/\sim=\{\frac{a}{b}: a\in R, b\in S\}$ where $\frac{a}{b}:=\{(a',b'):(a',b')\sim(a,b)\}$
Operations: $\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}$, $\frac{a}{b}\cdot\frac{c}{d}=\frac{ac}{bd}$, Identity: $0_{F_R}=\frac{0}{1}, 1_{F_R}=\frac{1}{1}$ Inverse: $-\frac{a}{b}=\frac{-a}{b}, (\frac{a}{b})^{-1}=\frac{b}{a}$ if $a\neq 0$.
Remark: $\tau: R\to F_R$ by $\tau(r)=\frac{r}{1}$ is a 1-1 ring homomorphism. ps:Isomorphism to $\{\frac{r}{1}: r\in R\}\subseteq F_R$. **$F_R$ is a field.**
Multiplicative Set: R communicative ring. $X\subseteq R$ is a *multiplicative set* if: **1**: $1_R\in X$ **2**: 乘法封闭 (option *3: $0_R\notin X$)
Ring of Fractions $(X^{-1}R)$**:** R 交换环, $X\subseteq R$ multiplicative set. Def *e.r.* $\sim$ on $R\times X$ by: $(a,x)\sim(b,y)\Leftrightarrow\exists s\in X$ s.t. $s(ay-bx)=0$.
Then the *ring of fractions* $X^{-1}R$: $X^{-1}R=(R\times X)/\sim=\{\frac{a}{x}: a\in R, x\in X\}$ where $\frac{a}{x}:=\{(a',x'):(a',x')\sim(a,x)\}$
Operations: $\frac{a}{x}+\frac{b}{y}=\frac{ay+bx}{xy}$, $\frac{a}{x}\cdot\frac{b}{y}=\frac{ab}{xy}$, Identity: $0_{X^{-1}R}=\frac{0}{1}, 1_{X^{-1}R}=\frac{1}{1}$ Inverse: $-\frac{a}{x}=\frac{-a}{x}$ If $a\in X$, then $(\frac{a}{x})^{-1}=\frac{x}{a}$.
Remark: $\varphi: R\to X^{-1}R$ by $\tau(r)=\frac{r}{1}$ is a ring homomorphism. ps:Isomorphism to $\{\frac{r}{1}: r\in R\}\subseteq X^{-1}R$. **$X^{-1}R$ is a ring.**
Localisation: P prime ideal of A. $\Rightarrow$ [1] $S=A\setminus P$ is a *multiplicative set*. [2] *Localisation of A at P*: $A_P=S^{-1}A$

Theorem: A communicative ring, then: $A_p=S^{-1}A$ has **only one maximal ideal**, which is $S^{-1}P$. ps: $S=A\setminus P$

5 Nullstellensatz, Variety & Vanishing Ideal

5.1 Nil and Jacobson Radical Ideals, Local Ring

Nil Ideal: I ideal of R is a *nil ideal* if: $\forall x \in I, \exists n \in \mathbb{N}$ s.t. $x^n = 0$. i.e. every element of I is nilpotent.

Nil Radical ($N(R) = \sqrt{0_R}$): $N(R) :=$ the largest nil ideal in R $\Leftrightarrow$ $N(R) :=$ sum of all nil ideals in R $\Leftrightarrow$ $N(R) = \sqrt{0_R}$

Note: sum of any two nil ideals is also a nil ideal. **Note**: $R/N(R)$ has no non-zero nilpotent elements.

Quasi-invertible: R ring, $a \in R$ is *quasi-invertible* if: $\exists b \in R$ s.t. $(1-a)(1-b) = 1$ $\leftrightarrow$ $ab = a + b$. (如果 a quasi-invertible, 那么 b 唯一)

Jacobson Radical Ideal: I ideal of R is a *Jacobson radical ideal* if: $\forall i \in I$, i is *quasi-invertible* in R.

Jacobson Radical $J(R)$: $J(R) :=$ the largest JRI in R $\Leftrightarrow$ $J(R) :=$ sum of all JRI in R $\Leftrightarrow$ $J(R) = \bigcap_{\text{(left)}}$ **maximal ideals of R.**

Note: sum of any two JRI is also a JRI. **Note**: **Nil ideal $\Rightarrow$ Jacobson Radical Ideal (JRI).**(通过等比数列证明)

Local Ring: R is a *local ring* if: R has **unique maximal ideal** $\Leftrightarrow R/J(R)$ is a field A_P **is a local ring.**(A commutative ring, P prime ideal of A)

Remark: $R/(JRI)$ or $R/J(R)$ has no non-zero nilpotent elements $(1-a)(1-b) = 1 \Leftrightarrow 1-a$ **is invertible** $(b = 1-(1-a)^{-1})$

5.2 Variety & Vanishing Ideal

Variety: Let k be a field, $f_1, \ldots, f_m \in k[x_1, \ldots, x_n]$. Then: **Variety:** $\mathbf{V}(f_1, \ldots, f_m) := \{(a_1, \ldots, a_n) \in k^n : f_i(a_1, \ldots, a_n) = 0, \forall i\}$

Vanishing Ideal: Let k be a field, $X \subseteq k^n$. Then: **Vanishing Ideal:** $\mathcal{I}(X) := \{f \in k[x_1, \ldots, x_n] : f(a_1, \ldots, a_n) = 0, \forall (a_1, \ldots, a_n) \in X\}$

Useful Tools: If $X = X_1 \cup X_2 \cup \cdots \cup X_m$, then $\mathcal{I}(X) = \mathcal{I}(X_1) \cap \mathcal{I}(X_2) \cap \cdots \cap \mathcal{I}(X_m)$ · If $I_1 \subseteq I_2$ ideals, then $\mathbf{V}(I_2) \subseteq \mathbf{V}(I_1)$

5.3 Nullstellensatz

Weak Nullstellensatz: $\mathbb{F}$ field, and $I \lhd \mathbb{F}[x_1, \ldots, x_n]$ ideal. Then: $(\Rightarrow)$ If $I = \langle x_1 - a_1, \ldots, x_n - a_n \rangle$, $a_i \in \mathbb{F}$, then I is maximal ideal.

Weak Nullstellensatz $(\Leftarrow)$ If I is *maximal ideal* and $\mathbb{F}$ is algebraically closed, then $I = \langle x_1 - a_1, \ldots, x_n - a_n \rangle$ for some $a_i \in \mathbb{F}$.

Fact I: If $I = \langle x_1 - a_1, \ldots, x_n - a_n \rangle$, then **[** $f(a_1, \ldots, a_n) = 0$ $\Leftrightarrow$ $f \in I$ **]**

Fact II: k field, $(\Rightarrow)$ I maximal ideal then $\mathbf{V}(I)$ **is a *single point*.** (i.e. $\mathbf{V}(I) = \{(a_1, \ldots, a_n)\}$) $(\Leftarrow)$ **if k algebraically closed.** (ps: 最好证明一下)

Reformation of Weak Nullstellensatz: k *algebraically closed* field, J ideals of $k[x_1, \ldots, x_n]$. Then:

[1] $\mathbf{V}(J) \neq \emptyset \Leftrightarrow J \neq k[x_1, \ldots, x_n]$ [2] $f \in \mathcal{I}(\mathbf{V}(J))$ $\Leftrightarrow$ $f^n \in J$ **for some** $n \in \mathbb{N}$ i.e. $\sqrt{J} = \mathcal{I}(\mathbf{V}(J))$ It's Strong Nullstellensatz!

Reformation of Hilbert's Nullstellensatz: k *algebraically closed* field, I ideal of $k[x_1, \ldots, x_n]$. Then:

[1] $\sqrt{I} = \bigcap$ **(maximal ideals $\supseteq I$)** $\Leftrightarrow$ [2] $J(R/I) = \bigcap$ **maximal ideals in R/I = nil radical of R/I** $= \sqrt{0_{R/I}}$

2nd Def of Radical: R commutative ring, I ideal of R. Then: $\sqrt{I} = \bigcap$ **(prime ideals $\supseteq I$)**

Corollary: If R communicative ring, then **Nil radical** $\sqrt{0_R} = N(R) = \bigcap$ **prime ideals of R =: Prime Radical.** If R 非交换, 不一定.

6 Gröbner Bases

Monomials: A *monomial* is a product of elements from $\{x_1, x_2, \ldots, x_n\}$. i.e. $f = x_1^{\alpha_1} x_2^{\alpha_2} \ldots x_n^{\alpha_n}$ (if 交换) $\deg(f) = \sum_i \alpha_i$

Lex Order (Lexicographic Order): [1] 定义变量顺序; [2] 看第一个不同的变量 e.g. $x > y > z$, then $x > yxxx$; $xy > yx$ (If NOT 交换)

Deglex Order: For f, g monomials with variables $x_1, x_2, \ldots, x_n$: [1] Compare degree; [2] If equal, define ordering on variables (e.g. $x_1 > x_2 > \ldots > x_n$) and compare lexicographically. If NOT 交换: $xxy > yxx$

multideg|LC|LM|LT & Ov: 分别为: 排序后的最大的一项的: 指数向量 | 系数 | 单项式 (coeff=1)| 整项 (完整的) **Overlap (Ov)**: f, g *poly*, $Ov(f,g) = \gcd(LM(f), LM(g))$

S-polynomial: For polynomials f, g: $S(f,g) = \frac{\mathbf{lcm}(LM(f), LM(g))}{LT(f)} f - \frac{\mathbf{lcm}(LM(f), LM(g))}{LT(g)} g =_{\text{or}} \frac{LT(g)}{Ov(f,g)} \cdot f - \frac{LT(f)}{Ov(f,g)} \cdot g$ 效果: f, g 的 LM 减小

Gröbner Basis: I ideal of $k[x_1, \ldots, x_n]$, k field. A subset $G \subseteq I$ is a *Gröbner basis* of I with respect to a *monomial order* (e.g. Lex, Deglex) if:

$\forall f \in I, f \neq 0$, $\exists g \in G$ s.t. $LM(g) \mid LM(f)$.

$(\Leftrightarrow)$ For any w_i *monomials*, $\alpha \in k$, s.t. **if** $LM(g) \nmid w_i$ **for all** $g \in G$**, then** $\sum_i \alpha_i w_i \notin I$.

Algorithm to Find Gröbner Basis: For ideal $I = \langle f_1, f_2, \ldots, f_m \rangle \lhd k[x_1, \ldots, x_n]$, k field. e.g. Buchberger's Algorithm

1. Let $G = \{f_1, f_2, \ldots, f_m\}$
2. For each $g_i \in G$, write $g_i = LC(g_i) \cdot LM(g_i) + h_i$
3. For each pair (g_i, g_j) in G, calculate $S(g_i, g_j)$ only $i > j$ or $i < j$ to avoid repetition **Remark**: If $Ov(g_i, g_j) = 1$, then $S(g_i, g_j) = 0$, skip it.
4. Reduce each $S(g_i, g_j)$: If $LM(g_k)$ divides any term of $S(g_i, g_j)$, substitute $LM(g_k)$ by $-h_k$; until no $LM(g_k)$ divides any term.
5. If $S(g_i, g_j) \neq 0$ after reduction, add it's leading term to G and go to step 2. If all $S(g_i, g_j) = 0$, stop and output G.

Brönner Basis Application: k field, $I \lhd k[x_1, \ldots, x_n]$, G Gröbner basis of I. L leading terms of G. W set of monomials not divisible by any element of L. (i.e. don't contain any monomials in L as a *subword*) Then:

1. $\{w + I : w \in W\}$ is a **basis of** $k[x_1, \ldots, x_n]/I$ **as a vector space over** k.
2. For each $g_i \in G$, write $g_i = LC(g_i) \cdot LM(g_i) + h_i$, **If** $r \in k[x_1, \ldots, x_n]$ **can be reduced to** 0 **by** $g_i \to -h_i$**, then** $r \in I$.

7 Appendix

Zorn's Lemma: If a *partially ordered set* (i.e. $\exists \sim_{\leq}$) P s.t. [1] $\forall$ *chain* in P has *upper bound*. [2] $P \neq \emptyset$. $\Rightarrow$ P has at least one *maximal element*. (i.e. $\exists m \in P$ s.t. $\forall p \in P$, if $m \leq p$, then $m = p$)

2nd Def of Prime Ideal: R communicative ring, I ideal of R. Then: I **prime ideal** $\Leftrightarrow$ **[J, K ideals s.t. $JK \subseteq I \Rightarrow J \subseteq I$ or $K \subseteq I$]** $\Leftrightarrow$ **[$J \subsetneq I, K \subsetneq I \Rightarrow JK \subsetneq I$]**